MILITARY SPECIFICATION

ASSEMBLY, KLYSTRON POWER SUPPLY: 10548905

1. SCOPE

1.1 This specification covers a +5 volt dc board assembly which generates the klystron tube filament voltage. This board assembly is part of the klystron power supply used in the Receiver-Transmitter RT-660/VPS-2, part of the Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

F10548009	PRF Counter, Electronic
D10548905	Board Assembly
C9115C6006	Connector, HV Miniature
C10548899	Holder, Connector

PACKAGING DATA SHEET

10548905	Packaging of Board Assembly, Klystron Power Supply

(Copies of specifications, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The board assembly shall be manufactured in accordance with Drawing D10548905 and drawings pertaining thereto.

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3.2 General specifications. - Where applicable the board assembly shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishings

d. Part identification and marking

e. Workmanship

f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies

b. Wiring

c. Threaded fasteners

d. Marking

e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration.- The board assembly shall be mounted on the vibration table in normal service position. The board assembly shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 55 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. - The board assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the

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three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 ±1 milliseconds (ms). The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

3.3.3 Altitude.

3.3.3.1 Operating. - The board assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. - The board assembly shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. - The board assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40° F and +155° F.

3.3.4.2 Nonoperating (storage). - The board assembly shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65° F and +160° F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). - The board assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ±5 percent for a period of not less than 10 days.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the +5 volt dc board assembly.

3.5 Performance. - The input and output voltage requirements shall be as specified in Table I.

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TABLE I. - INPUT AND OUTPUT REQUIREMENTS

REQUIREMENT	CHARACTERISTICS	LOCATION
Input voltage 14 to 16 V peak-to-peak	PRF: 3.5 ±0.1 kHz square wave 50% duty cycle.	See Figure 1
Output voltage 5.8 Vdc	5.8 ±0.3 Vdc. When the board assembly output terminals are connected to the test fixture as shown in 4.5.2.3.	J2C(+) and J2B(-)

3.5.1 Output load current shall not be less than 2.5 amperes (A) and not more than 3.1 A when operating as connected in Figure 1.

3.5.2 Output ripple shall not exceed 100 millivolts (mV) peak-to-peak, excluding spikes when the output current is as stated in 3.5.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) submission. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three board assemblies manufactured in the same manner, using the same materials, equipment,

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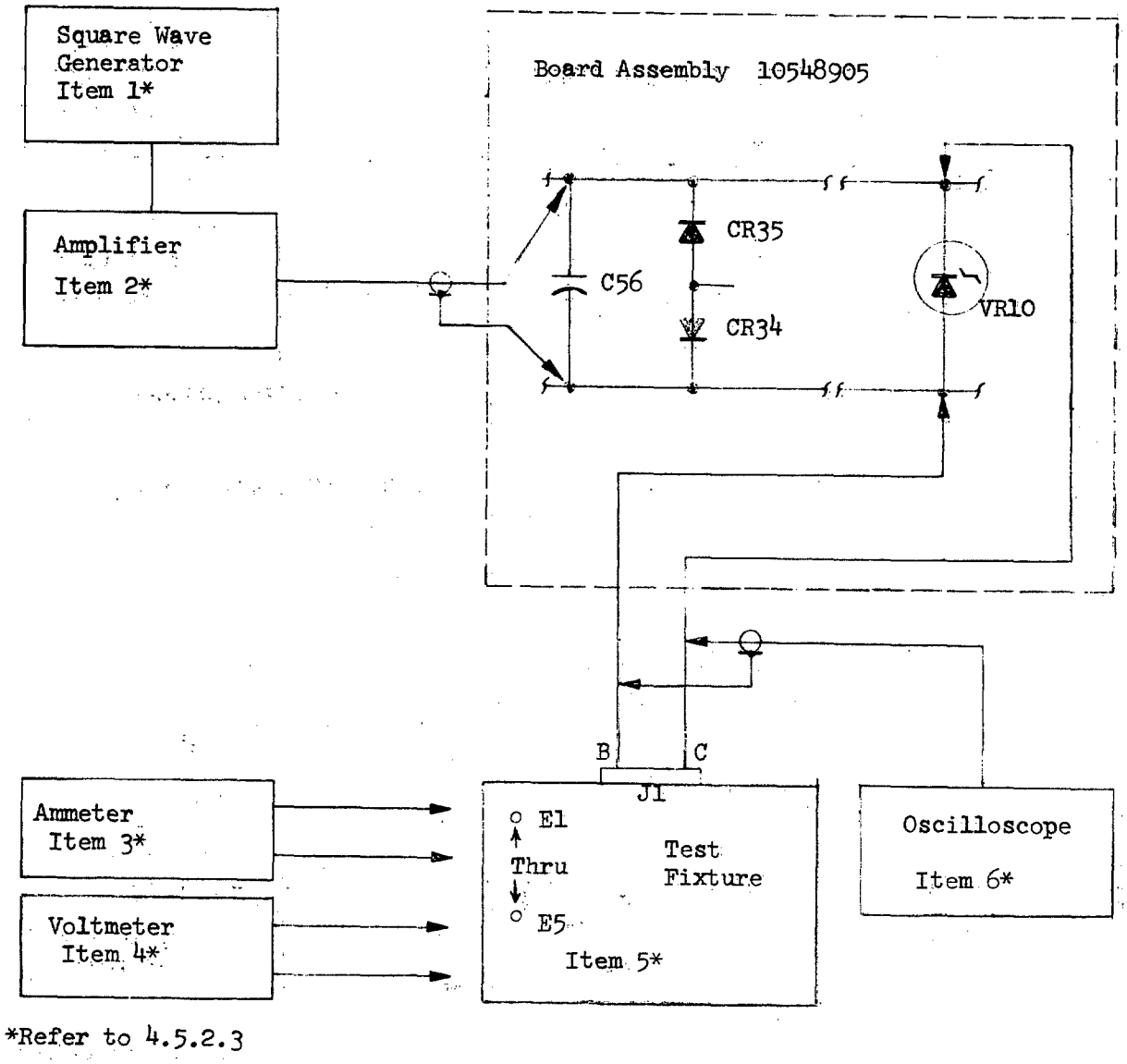


FIGURE 1. BOARD ASSEMBLY TEST SETUP.

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processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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4.3.2.3 Functional tests. - The requirements and tests in Table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60° F and 90° F).

TABLE II. - CLASSIFICATION OF DEFECTS

(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
MINOR: NONE		

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. - One board assembly shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table III. Three board assemblies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table IV.

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TABLE III. - SPECIAL SAMPLING

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings-Visual
302. General specifications	3.2	MIL-F-13926 & MIL-C-46357-Visual

TABLE IV. - SPECIAL SAMPLING, ENVIRONMENTAL

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303. Vibration	3.3.1	4.6.1.1
304. Shock	3.3.2	4.6.1.2
305. Altitude	3.3.3	4.6.1.3
306. Temperature	3.3.4	4.6.1.4
307. Humidity (nonoperating)	3.3.5	4.6.1.5

4.4.1 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

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4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment. - In conjunction with 4.5.2.2, the standard test equipment listed in Table V shall be utilized in performing the applicable tests of 4.6.

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TABLE V. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Square wave generator	Capable of producing 3.5 kHz, 14 to 16 V. Peak-to-peak square wave, 50% duty cycle.
2. Power amplifier	Power response 50 to 100 W at less than 5% distortion, +1 dB from 20 to 50,000 Hz. Output capable of delivering 20 V and 3.0 A.
3. Ammeter	Dc, 0-5 A, accuracy of 1%.
4. Voltmeter	Dc, 0-10 V, accuracy of 1%, 20,000 ohms/V.
5. Test fixture	As specified in Figure 2. See Table VI for parts list.
6. Oscilloscope	Input sensitivity: 10 mV/division. Input impedance: 1 megohm.

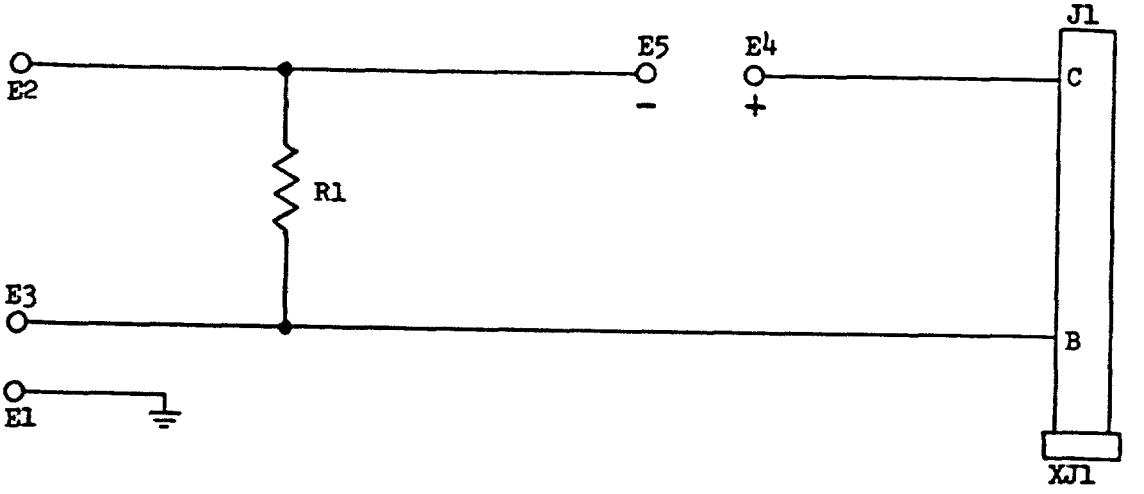


FIGURE 2. TEST FIXTURE SETUP.

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TABLE VI. - PARTS LIST FOR TEST FIXTURE

DESIGNATION	PART	DESCRIPTION
E1	E. F. Johnson 111-103	Binding post (black)
E2, E4, E5	E. F. Johnson 111-102	Binding post (red)
E3	E. F. Johnson 111-104	Binding post (green)
J1	9115C6006	Connector, HV miniature
R1		Resistor; wirewound, adjustable, 3 ohm, 50 W
XJ1	10548899	Holder, connector

4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I of MIL-STD-810 except that failure criteria do not apply. After vibration, the board assembly shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Shock. - This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 G. After the shock impulses, the board assembly shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.2.

4.6.1.3 Altitude (nonoperating and operating). - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 feet (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

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4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810. The temperature of Step 2 shall be +160° F. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +155° F. In Step 5, the temperature shall be maintained for 4 hours and the board assembly shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

4.6.1.4.2 Low (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -65° F maintained for not less than 4 hours. The temperature of Step 4 shall be -40° F. In Steps 5 and 7, the board assembly shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity (nonoperating). - This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The board assembly shall be tested in accordance with 4.6.3.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.6.3 Performance. - The selected board assemblies shall be tested as follows: All following tests shall be performed using the test equipment described in Table V and as connected in Figure 1.

4.6.3.1 Output voltage. - Using square-wave generator, item 1, apply 16-V peak-to-peak square wave in accordance with Table I and Figure 1. Using voltmeter, item 4, verify that the output voltage at E2 (-) and E3 (+) is as specified in Table I. Using ammeter, item 3, verify load current between E5 (-) and E4 (-) is as specified in 3.5.1.

4.6.3.2 Output ripple. - Using oscilloscope, item 6, verify that the output ripple between J1-C and J1-B (common) is as specified in 3.5.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548905. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use. - The board assembly rectifies and filters the 5 Vdc filament voltage required for functional operation of the klystron tube amplifier. The filament voltage is raised to the high voltage (9-kilovolt (kV)) level of the klystron tube cathode. The board assembly is part of the klystron power supply, 2A2 unit, used in the Radar Set AN/VPS-2.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A258

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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